Ding Yan(丁琰)

I am an aggressive young student loving solving issues and creating unexplored values for people. I am the only one **Academic Star** (master, 2018) at the College of Computer Science in Chongqing University, China, as well as the only one **Academic Star** (2019) at BDSC group. Besides, I am the only one **HUAWEI First-Class Scholarship** (master, 2018) at Chongqing University, China. I have completed eight papers as the first author, including PerCom, IEEE TITS, TII and TKDE in the last four years.

Education Background

2016.09 - 2019.06 (Expected), Chongqing University, China

M.S, College of Computer Science, majoring in Computer Science and Technology
Supervised by Full Prof. Chao Chen, GPA: 80.03/100

2014.09 - 2016.06, Chongqing University, China (Top 15 in Mechanical Engineering)

B.E, College of Mechanical Engineering, majoring in Mechatronics Engineering
GPA: 3.23/4, Ranking: 9/125

2012.09 - 2014.06, Chongqing University, China (Top 5 in Vehicle Engineering)

B.E, College of Mechanical Engineering, majoring in Vehicle Engineering

Research Directions

Intelligent Transport & Pervasive Computing & Urban Computing & Spatial-temporal Trajectory Data Mining

Publications

Conference Paper

- [1] **Yan Ding**, Chao Chen, et al. *GreenPlanner: Planning personalized fuel-efficient driving routes using multi-sourced urban data*. **IEEE International Conference on Pervasive Computing and Communications (PerCom)**, Pages 207-216, Hawaii, USA, March 13-17, 2017. (CCF-B, accepted rate:14.8%)
- [2] **Yan Ding**, Chao Chen, et al. *Fuel Consumption Estimation of Potential Driving Paths by Leveraging Online Route APIs*, **The 13th International Conference on Green, Pervasive and Cloud Computing (GPC)**, Pages X-Y, Hangzhou, China, April 11-14, 2018. (Accepted, EI-index).
- [3] **Yan Ding**, Chao Chen, et al. *An Online Trajectory Compression System Applied to Resource-Constrained GPS Devices in Vehicles*, **IEEE International Conference on Sensing, Communication and Networking (SECON Workshops)**, HongKong, China, June 11-13, 2018. (CCF-B).

Journal Paper (#equal contribution)

- [4] Chao Chen#, **Yan Ding#**, et al. *A Novel Online Trajectory Compression Framework Leveraging Vehicle Heading Direction and Change.* In: **IEEE Transactions on Intelligent Transportation Systems, 2018.** (Accepted, IF:4.07).
- [5] Chao Chen#, **Yan Ding#**, et al. *A Three-stage Online Map-Matching Algorithm by Fully Using Vehicle Heading Direction.* In: **Journal of Ambient Intelligence and Humanized Computing, 2018.** (Accepted, CCF-C, IF:1.588).
- [6] Chao Chen#, **Yan Ding#**, et al. *VTracker: When Online Trajectory Compression Meets Edge Computing.* In: **IEEE Transactions on Industrial Informatics, 2018.** (Major Revision, IF:6.764).
- [7] **Yan Ding**, et al. *Eco-Route: Recommending Economical Driving Routes For Plug-in Hybrid Electric Vehicles.* (arXiv).
- [8] Yuanjian Zhang, Chu Liang, **Yan Ding**, et al. *A Hierarchical Energy Management Strategy Based on Model Predictive Control for Plug-in Hybrid Electric Vehicle.* In: **IEEE Transactions on Vehicular Technology, 2018.** (Submitted in 2018.06, IF:4.432).

- [9] **Yan Ding#**, Chao Chen#, et al. DAVT: A Nearly-All-Information-Preserving Data Representation and Compression Framework for Temporal Trajectories. In: IEEE Transactions on Knowledge and Data Engineering. (Submitted in 2019.02, IF: 2.775)

#CCF A/B/C is a significant academic standard proposed in China to evaluate the quality of the journals and conferences in the computer science field, and only a few are on the list for recommendation.

Activities Experience

- On April 11-14, 2018, oral presentation (25 minutes) in Hangzhou, China
- On March 13-17, 2017, oral presentation (20 minutes) in Hawaii, USA.

Awards & Honors

- **Huawei First-Class Scholarship (one master student per year at Chongqing University, 2018)**
- **Academic Star candidate at the College of Computer Science (one person per year at the college, 2018)**.
- **Academic Star in BDSC group (one person per year at the group, 2019)**.
- Second-Class Scholarship of Chonging University (top 10%, 2013).
- Third-Class Scholarship of Chonging University (top 15%, 2013).
- Third-Class Scholarship of Chonging University (top 15%, 2014).
- National Training Programs of Innovation for Undergraduate (2014).
- Student Research Training Program (2014).
- Mathematical Contest In Modeling (Honorable Mention, 2016).

Patents

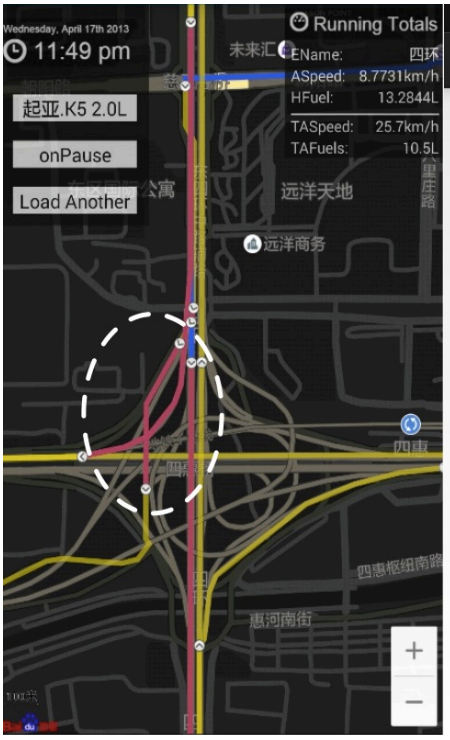
- Chao Chen, **Yan Ding** and Xuefeng Xie, CN 108253976 A (2018). *The patent is based on the paper* [5].
- Chao Chen, **Yan Ding**, Liang Feng, Yun Lin and Yandong He, CN 107014389 A (2017) . *The patent is based on the paper* [1].
- **Yan Ding** and Chao Chen. *The patent aims to plan safe and fuel-efficient flight lines when unmanned aerial vehicles deliver pint-sized goods (The patent has already been submitted)*.
- Yuan Wang, Yun Kong, Rui Yan and **Yan Ding**, CN 203936094 U (2014). The patent is to design a bottle washer.

Application Demos

- **FuelVisualizer** (left figure) is to *display the real-time fuel-consumption of vehicles* on paths in the city of Beijing, China. Fuel consumption is classified into three types i.e.,high, medium and low in terms of the fuel consumption per hundred kilometers. Types are represented by three colors, and more specifically blue, yellow and red refer to low, medium and high fuel usage, respectively. This demo is based on the paper [1].
  Project Period: 2015.9∼2016.6

- **GreenAPI** (middle figure) is to *recommend the most fuel-efficient driving route* for users among all routes from the beginning to the destination provided by online route APIs (e.g., Gaode Map). The demo is based on the two papers [1][2].
  Project Period: 2017.12∼2018.6

- **TrajCompressor** (right figure) is to *online compress the raw GPS trajectory data* to reduce the storage and communication cost. The demo is based on the three papers [4][5][6]
  Project Period: 2017.9∼2018.5



FuelVisualizer

GreenAPI

TrajCompressor

Skills

Programming

MATLAB(Skillful), Python(Basic), Java(Basic), LaTeX(Skillful) and C/C++(Basic)

Language

TOFEL (83 scores): Reading 23, Listening 19, Speaking 19, Writing 22
GRE (319+3.5 scores): Verbal 151 (52nd), Quantitative 166 (90th), Writing 3.5 (41st)